

Reckitt Sales Analytics Pipeline & Dashboard

Overview

The Reckitt FMCG Sales Analytics project implements an end-to-end data engineering and business intelligence workflow that transforms raw FMCG sales data into an interactive Power BI dashboard.

The final dashboard provides a consolidated view of **sales performance across time, products, brands, geography, store types, suppliers, warehouses, inventory, and order status.**

Dashboard Objectives

- Build a reproducible pipeline for processing raw FMCG sales data.
 - Validate and clean data before it reaches the reporting layer.
 - Derive additional analytical features from the source data.
 - Transform the flat source dataset into a Power BI–optimized star schema.
 - Develop reusable business KPIs using DAX.
 - Provide interactive analysis across sales, product, geography and supply-chain dimensions.
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Data Pipeline

Data Ingestion

The raw monthly sales CSV is ingested using **PySpark** with an explicitly defined schema.

The ingestion layer:

- Reads the raw source data.
- Enforces expected column names and data types.
- Normalizes column-name whitespace.
- Produces a Spark DataFrame for downstream processing.

Using an explicit schema provides a consistent data contract instead of relying on automatic schema inference.

Data Quality Validation

Data quality checks are performed to identify issues before transformation.

Validation includes:

- Row and column counts.
- Duplicate-record detection.
- Column-level null percentages.
- Configurable null-threshold checks.
- Identification of data-quality issues requiring cleaning.

For the current dataset:

Raw Records: 11,271

Duplicate Records: 221

Records after Deduplication: 11,050

This validation stage ensures that known quality problems are identified before the data reaches the analytical model.

Data Cleaning

The cleaning layer standardizes the source data while preserving valid business records.

Key operations include:

- Row de-duplication
- Handling null/missing entries
- Conversion of invalid numeric placeholders to `NULL` before numeric casting.
- Standardization of fields required by downstream transformations.

The cleaned dataset becomes the base for feature engineering and dimensional modeling.

Feature Engineering

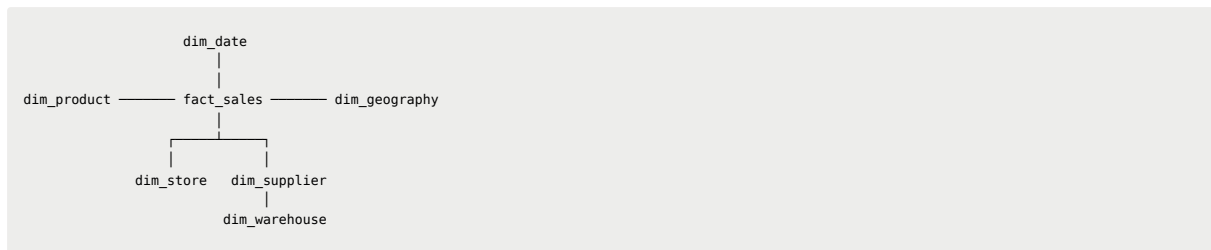
Additional analytical fields are derived from the cleaned source data. These fields extend the raw source data with attributes required for downstream business analysis.

Derived Feature	Purpose
Quarter	Enables quarterly analysis
YearMonthKey	Provides chronological Year-Month ordering
PricePerUnit	Transaction-level selling-price indicator
CategorySalesRank	Ranks sales performance within categories
InventoryRiskFlag	Identifies records requiring inventory attention

Data Modeling

Data Model

The Power BI semantic model follows a **star schema**, with **fact_sales** as the central fact table.



Fact Table **fact_sales** - Contains the measurable sales and operational information used by dashboard calculations, including:

- Value Sales
- Units Sales
- Supplier Lead Time
- Inventory Status
- Order Status
- Price Per Unit
- Inventory Risk Flag

Dimension Tables

Dimension	Purpose
dim_date	Year, month, quarter and reporting-period analysis
dim_product	Product, category, segment and brand analysis
dim_geography	Region, state and city analysis
dim_store	Store and store-type analysis
dim_supplier	Supplier-level analysis
dim_warehouse	Warehouse-level analysis

Key Performance Indicators (KPIs)

KPI	Purpose
Total Revenue	Overall sales value generated
Total Units Sold	Total sales volume
Average Selling Price	Revenue generated per unit sold
Net Revenue	Revenue after applying the defined valid-order logic
Revenue YoY %	Year-over-year revenue change
Revenue MoM %	Month-over-month revenue change
Revenue YTD	Cumulative revenue within the selected year
Average Lead Time	Average supplier lead time
Brand Revenue Rank	Revenue-based ranking of brands

DAX Measures

A dedicated Measures table is used to organize DAX calculations separately from fact and dimension columns.

1. **Total Revenue** - Provides the base revenue measure used throughout the report.

```
Total Revenue =
SUM ( fact_sales[value_sales] )
```

2. **Total Units Sold** - Measures sales volume under the current filter context.

```
Total Units Sold =
SUM ( fact_sales[units_sales] )
```

3. **Average Selling Price** - Uses total revenue divided by total units rather than averaging row-level `PricePerUnit`, providing the correct weighted selling price.

```
Average Selling Price =
DIVIDE (
    [Total Revenue],
    [Total Units Sold]
)
```

4. **Revenue YoY %** - Compares current-period revenue against the corresponding prior-year period.

```
Revenue YoY % =
VAR CurrentRevenue = [Total Revenue]
VAR PriorRevenue = [Revenue PY]

RETURN
    DIVIDE (
        CurrentRevenue - PriorRevenue,
        PriorRevenue
    )
```

5. **Brand Revenue Rank** - Ranks brands by revenue while respecting the user's current report selections.

```
Brand Revenue Rank =
RANKX (
    ALLSELECTED ( dim_product[BRAND_2] ),
    CALCULATE ( [Total Revenue] ),
    ,
    DESC,
    DENSE
)
```

6. **Brand Contribution %** - Shows how much the selected brand contributes to overall revenue within the remaining filter context.

```
Brand Contribution % =
DIVIDE (
    [Total Revenue],
    CALCULATE (
        [Total Revenue],
        REMOVEFILTERS ( dim_product[BRAND_2] )
    )
)
```

DAX Concepts Applied

`CALCULATE` • `FILTER` • `VAR/RETURN` • `DIVIDE` • `SELECTEDVALUE` • `ALL` • `ALLSELECTED` • `REMOVEFILTERS` • `RANKX` • Filter Context • Time Intelligence

Dashboard Analysis

1. Executive / Sales Overview

Provides a high-level view of business performance through:

- Revenue
- Units Sold
- Average Selling Price
- Revenue growth
- Monthly revenue trends
- Category performance
- Regional performance

2. Product / Brand Analysis

Provides deeper analysis of product and brand performance through:

- Revenue by brand
- Units sold
- Category and segment performance
- Brand ranking
- Brand contribution
- Selling-price analysis
- Performance trends over time

- Brand performance
- Store-type performance

3. Geographic Analysis

Analyzes sales performance across:

- Region
- State
- City

This enables identification of strong and weak geographical markets.

4. Supply Chain Analysis

Uses the available operational data to analyze:

- Supplier lead time
- Supplier performance
- Inventory status
- Order status
- Warehouse performance
- Inventory risk indicators

Dashboard Design Approach

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Executive KPIs
  ↓
Performance Trends
  ↓
Category / Brand / Geography Analysis
  ↓
Operational Analysis
  ↓
Drill-Through Details
  
```

Key Learnings

The project provided practical exposure to the complete flow between data engineering and business intelligence:

- Building a modular **ETL pipeline using PySpark**.
- Applying explicit schemas and data-quality validation.
- Cleaning and standardizing real-world data issues.
- Deriving analytical features through feature engineering.
- Understanding the importance of **natural-key uniqueness** when constructing dimensions.
- Designing a **star schema** and understanding fact/dimension table grain.
- Connecting curated engineering outputs to a Power BI semantic model.
- Creating and managing **one-to-many relationships and filter propagation**.
- Developing reusable DAX measures and understanding **row context vs filter context**.
- Implementing ranking, time-intelligence and context-aware calculations.
- Translating business requirements into appropriate KPIs and visualizations.
- Implementing slicers, cross-filtering, drill-through, tooltips and bookmarks.
- Understanding how **data engineering, dimensional modeling and BI visualization form a single end-to-end analytical workflow**.

Overall Flow

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Raw FMCG CSV
  ↓
PySpark Ingestion
  ↓
Data Quality Validation
  ↓
Data Cleaning
  ↓
Feature Engineering
  ↓
Star Schema Construction
  ↓
Curated CSVs
  ↓
Power BI Import
  ↓
Fact-Dimension Relationships
  ↓
DAX Measures & KPIs
  
```

↓
Analytical Visualizations
↓
Interactive BI Dashboard

Outcome: An end-to-end analytics solution that transforms raw FMCG sales data into a validated, structured analytical model and an interactive Power BI dashboard for sales, product, geographic and supply-chain analysis.

References

GitHub Repository

| <https://github.com/uthrahh/Reckitt-Dashboard>

Dataset

| <https://www.kaggle.com/datasets/agnimchakraborty/reckitt>